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A practitioner's guide to isometric hip adduction and abduction test selection: Maximal force, asymmetry, and muscle activity comparisons between assessment positions

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ABSTRACT

The purpose of this study was to compare maximal adduction (ADD) and abduction (ABD) force, ADD to ABD ratio (ADD:ABD), inter-limb asymmetries and muscle activity between five isometric hip strength assessment positions. Twenty male athletes performed the following positions: seated (SE), supine at knees with 0° hip flexion ($SK_{0°}$) and 45° hip flexion ($SK_{45°}$), and supine at ankles bilaterally (SA_{BI}) and unilaterally (SA_{UNI}). Normalised muscle activity ($\%EMG_{max}$) of adductor longus (ADD_{LONG}), gracilis (GRAC), gluteus medius (G_{MED}), sartorius (SAR), and lower rectus abdominis (REC_{ABD}) were also assessed. Significant main effects of assessment position were identified for relative peak force (rPF) and torque (rT), ADD:ABD, ADD asymmetry, and $\%EMG_{max}$. Significantly greater ADD and ABD rPF and rT were observed in SE and $SK_{45°}$ compared to SA_{BI} and SA_{UNI}. $SK_{45°}$ elicited the highest ADD_{LONG} and GRAC $\%EMG_{max}$ during ADD, with SE and $SK_{45°}$ evoking the highest G_{MED} $\%EMG_{max}$ during ABD. $SK_{0°}$ exhibited significantly higher ADD:ABD compared to all positions except SA_{UNI}. ADD asymmetry was significantly largest in SA_{UNI}, and REC_{ABD} $\%EMG_{max}$ was highest in SA_{BI} during ADD. This data can inform selection of assessment positions based on whether the primary measures of interest are maximal force and muscle activity, or identifying asymmetries.

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Electromyography; groin; rectus abdominis; strength; torque; reliability

1. Introduction

Isometric strength assessments of individual joint actions are widely employed by practitioners in high-performance sports and clinical settings (Herrington et al., 2018). These assessments allow for initial benchmarking of an athlete or client's performance against normative values, and subsequent monitoring against their own data (Wilson & Murphy, 1996). This enables practitioners to identify specific joint-angle weaknesses or muscular strength imbalances that may increase injury risk or restrict athletic performance (Kollock et al., 2015). Provided that appropriate assessments are selected and the data is effectively interpreted, evidence-informed decisions can be made regarding training planning, with targeted exercise interventions prescribed to reduce the opportunity for injury, improve technical performance, or guide rehabilitation (Herrington et al., 2018; Thorborg et al., 2011).

Assessments of maximal isometric hip adduction (ADD) and abduction (ABD) strength are becoming increasingly common in these environments due to reported associations with lower-limb injuries. Specifically, previous research has identified significant relationships between heightened risks of non-contact groin, hip, and knee injuries, with factors such as an ADD to ABD strength ratio (ADD:ABD) of less than 1.0, low ADD strength, and an inter-limb asymmetry for ADD and ABD strength (Bourne et al., 2020; Collings et al., 2022; De Blaiser et al., 2021; Markovic et al., 2020; Thorborg et al., 2011; Tyler et al., 2001). With the increased availability of testing devices consisting of a rigid frame and externally fixated load cells, recognised as the gold standard for these assessments, an extended range of reliable assessment positions are now available for selection by practitioners (O'Brien et al., 2019; O'Connor et al., 2023). These positions can involve the athlete or client being supine or seated, applying force bilaterally or unilaterally at the knee (short-lever) or ankle (long-lever), and adopting various joint-angle combinations, including 0°, 45°, 60° and 90° for the hip and knee joints. However, no research to date has directly compared the maximal force or asymmetry measures between these assessment positions on such testing devices.

Whilst there is clear value to incorporating isometric hip ADD and ABD strength assessments in screening and monitoring protocols, large variation exists in the positions utilised in high-performance environments, sports medicine clinics, and research studies (Collings et al., 2022; O'Brien et al., 2019; O'Connor et al., 2023). Notably, significant population variability exists in lower-limb length ratios, and each hip adductor and abductor muscle exhibits distinct attachment sites, orientation, and muscle architecture (Holliday & Ruff, 2001; Ward et al., 2010). This anatomical diversity suggests that substantial differences are likely to exist between the various assessment positions for maximal force, ADD:ABD, the inter-limb asymmetries, and activity of the adductor and abductor muscles. Consequently, the

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previously identified relationships between lower-limb injury risk and these measures in one assessment position may not directly translate to others, impacting test selection and data interpretation. Additionally, given the frequent use of these assessments in sports medicine for rehabilitation monitoring, knowledge of the assessment positions that elicit the greatest muscle activity for the commonly injured muscles is crucial (Thorborg et al., 2011). Therefore, the purpose of this study was to compare the maximal ADD and ABD strength, ADD:ABD, inter-limb asymmetries, and muscle activity across five commonly utilised hip ADD and ABD assessment positions. It was hypothesized that these variables would exhibit significant differences between the assessment positions due to the considerable variation in joint-angle combinations and points of force application.

2. Materials and methods

2.1 Study design

To compare the maximal force output, ADD:ABD, inter-limb asymmetries, and muscle activity in five commonly utilised isometric hip ADD and ABD assessments, athletes performed the assessments during a single session in a randomised order, to account for order effect. The assessment positions included: seated with 90° of hip and knee flexion (SE), supine at the knees with 0° knee and hip flexion ($SK_{0°}$), supine at the knees with 45° hip flexion ($SK_{45°}$), supine at the ankles with 0° hip and knee flexion and bilateral force application (SA_{BI}), and supine at the ankle with 0° hip and knee flexion and unilateral force application (SA_{UNI}) (Figure 1). During all assessments, the athletes' magnitude of muscle activity on each side was recorded with surface electromyography (EMG) for the adductor longus (ADD_{LONG}), gracilis (GRAC), gluteus medius (G_{MED}), sartorius (SAR), and lower rectus abdominis (REC_{ABD}). Additionally, prior to the ADD and ABD assessments athletes performed three trials of a resisted adduction sit-up test (RASUT) to provide the maximal EMG amplitude (EMG_{max}) for the REC_{ABD} (Galano et al., 2021).

2.2. Subjects

Twenty male ice hockey athletes (28.2 ± 5.9 yrs; 84.5 ± 8.8 kg; 179.8 ± 7.2 cm) volunteered to participate in this study. Athletes were recruited from a single team competing in a national-level competition (Australian Ice Hockey League), all of whom had a minimum of 10 years of experience in the sport and at least four years of resistance training prescribed by an accredited strength and conditioning coach. Testing occurred during pre-season training and athletes were required to be free of any pre-existing medical conditions that would restrict their ability to complete all testing. An *a priori* estimation revealed that to achieve a minimum power of 0.80 with a moderate effect (0.3) for a within-subjects comparison of the five assessment positions, a sample size of 20 athletes were required. This calculation was in agreement with previous research that employed a similar research methodology and design (Serner et al., 2014). Ethical approval was obtained through the University of Newcastle Human Ethics Committee (H-2022–0297), and information about the study was provided to all athletes prior to providing informed consent and undertaking screening for medical contraindications.

2.3. Measures

Five isometric hip ADD and ABD assessment positions were performed on a purpose-built device, consisting of a rigid frame and four independent uniaxial load cells (ForceFrame, VALD Performance, Brisbane, Australia). The device was interfaced with a tablet (iPad 10th Gen.; Apple Inc., Cupertino, USA) and sampled at 100 hz (Bourne et al., 2020; Secomb et al., 2021). To prepare for the maximal effort testing, participants performed the RASUT, followed by two submaximal efforts (60% and 80% of maximum, respectively) in their first randomly allocated assessment position (O'Brien et al., 2019). The bar height was adjusted for each assessment position to ensure that the athletes could assume the required hip and knee joint angles, which were confirmed with an extended-arm goniometer. Additionally, for the SE position the bar height was adjusted to ensure that the entire sole of both feet were in contact with the ground, whilst maintaining 90° of hip and

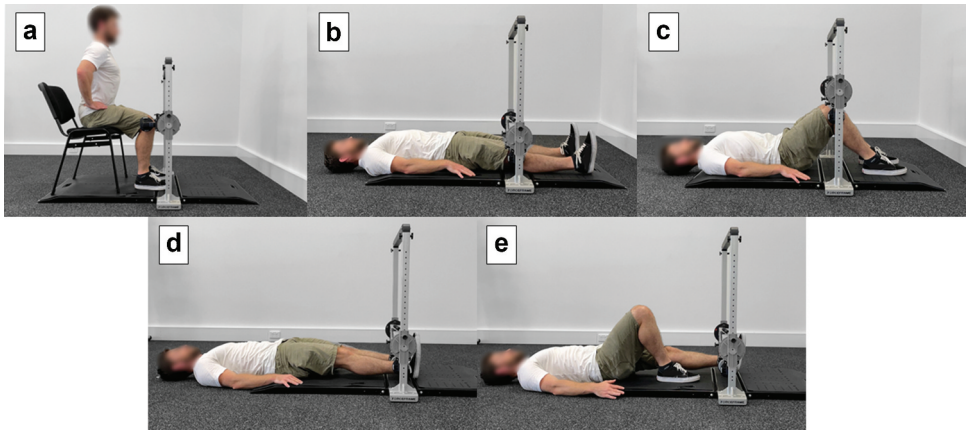

Figure 1. The isometric hip adduction and abduction assessment positions. (a) seated with 90° of hip and knee flexion [SE], (b) supine at the knees with 0° knee and hip flexion [$SK_{0°}$], (c) supine at the knees with 45° hip flexion [$SK_{45°}$], (d) supine at the ankles with 0° hip and knee flexion and bilateral force application [SA_{BI}], and (e) supine at the ankle with 0° hip and knee flexion and unilateral force application [SA_{UNI}].

knee flexion. Furthermore, for the SA_{UNI} position the athletes were required to press their arms and support (non-assessed) leg into the ground to ensure that no trunk movement occurred during assessment (Secomb et al., 2023).

For the maximal effort trials, athletes were provided with approximately 4 s to achieve peak force (PF), and the verbal instructions to "squeeze as hard as possible" for the ADD trials, and to "push as hard as possible" against the load cells for the ABD trials (Secomb et al., 2023). Two maximal effort trials were performed in each position and separated by 90 s. The trial that elicited the greatest PF in each assessment position was used for further analysis, with additional calculations performed to provide maximal relative isometric force (relative PF [rPF]; PF divided by the athlete's BW) and maximal relative rotational force (relative torque [rT]; PF multiplied by the associated lower-limb length, and divided by the athlete's BW). Lower-limb lengths were recorded as the distance (in metres) from the pubic tubercle to the medial midline of the knee for the SE, $SK_{0°}$ and $SK_{45°}$ positions, and from the pubic tubercle to the medial malleolus for SA_{BI} and SA_{UNI} positions. To calculate the inter-limb asymmetries for ADD, ABD and ADD:ABD, the symmetry angle (SA; % = (((45° - arctan[left leg/right leg])/90°) × 100)) was utilised as this method negates the requirement for subjective selection of a limb as the reference value, which can artificially inflate the asymmetry calculation (Zifchock et al., 2008). Furthermore, the intra-session reliability of the two maximal effort trials were assessed using two-way, fixed effects intra-class correlation coefficients for absolute agreement and the coefficient of variation (CV%), with results presented in Table 1.

Muscle activity was recorded with surface EMG electrodes sampling at 2000 hz (Trigno Avanti Research Plus; Delsys, Boston, USA) during all trials, including the RASUT. The sites were prepared, abraded, and cleaned with alcohol rubbing wipes in accordance with SENIAM recommendations, and the electrodes fixated with double-sided adhesive tape (Lovell et al., 2012). To determine sensor locations the following protocols were used: ADD_{LONG}; 40% of the distance from the pubic tubercle to the medial midline of the knee, *GRAC;* 20% of the distance from the pubic tubercle to the medial midline of the knee, G_{MED}; 50% of the distance between iliac crest and greater trochanter, *SAR;* 8 cm distal of the anterior superior iliac spine, and REC_{ABD}; 50% of the distance between umbilicus and pubic crest and 3 cm lateral to the midline (Elsais, 2019; Lovell et al., 2012; Ng et al., 1998; Serner et al., 2014). Electrodes were placed at least 3 cm apart from each other to avoid unstable recordings, and following initial mark-up, athletes performed voluntary contractions against manual resistance to confirm correct sensor placement on the muscle belly. During these voluntary contractions, the EMG signal qualities were checked, with the site cleaned again and sensor position adjusted if low-quality signals were detected.

All data were analysed with the proprietary software (EMGworks Analysis; Delsys, Boston, MA) following testing. EMG signals were rectified, using the *"absolute values"* function, prior to offline bandpass filtering between 10 and 500 hz using a Butterworth 2nd order filter. Once filtered, the root mean square (RMS) was calculated using an averaging window length of 500 ms and a window overlap of 250 ms, as previously reported as appropriate for maximal isometric strength assessments (Buckthorpe et al., 2012). The highest average EMG amplitude for each muscle was taken from the one-second period of the RMS data that corresponded with the PF from the best ADD and ABD trial (highest PF) in each assessment position. To normalise the EMG data relative to the individual's maximal recorded amplitude for each muscle, all RMS values were converted to a percentage of EMG_{max} ($\%EMG_{max}$). This was performed by setting the highest RMS value achieved by the athlete for each muscle across all five assessment positions as the $100\%EMG_{max}$ for that muscle. Subsequently, the data for that muscle in the other assessment positions were converted relative to this maximal value. For the bilateral assessments (SE, $SK_{0°}$, $SK_{45°}$ and SA_{BI}) the average of the left and right legs were used for analysis. Whereas for the unilateral assessment (SA_{UNI}) separate average calculations were made for both the assessed leg and support leg, due to the requirement to utilise the non-assessed leg for bracing. For example, the average of the $\%EMG_{max}$ on the left ADD_{LONG} from the SA_{UNI} left leg ADD trial and the right ADD_{LONG} from the SA_{UNI} right leg ADD trial, were averaged, and used as the ADD_{LONG} "assessed leg" value. Additionally, the $100\%EMG_{max}$ for the REC_{ABD} was set from the RASUT trial that elicited the highest EMG amplitude for the left and right side, with the two values averaged to provide the reference value for $\%EMG_{max}$ conversions.

Table 1. Intra-session reliability of the two maximal effort trials for adduction (ADD) and abduction (ABD) across the five assessment positions, with the coefficient of variation (CV%) and intra-class correlation coefficient (ICC) with 95% confidence intervals (95%CI). *SE = seated; $SK_{0°}$ = supine at the knees with 0° hip and knee flexion; $SK_{45°}$ = supine at the knees with 45° hip flexion; SA_{BI} = supine at the ankles bilaterally; SA_{UNI} = supine at ankle unilaterally.*

		CV%	ICC (95%CI)
SE	ADD	2.4	0.98 (0.97-0.99)
	ABD	2.3	0.99 (0.98-0.99)
$SK_{0°}$	ADD	2.4	0.99 (0.98-0.99)
	ABD	3.1	0.99 (0.98-0.99)
$SK_{45°}$	ADD	2.8	0.99 (0.98-0.99)
	ABD	2.0	0.99 (0.98-0.99)
SA_{BI}	ADD	3.3	0.97 (0.95-0.98)
	ABD	2.5	0.98 (0.97-0.98)
SA_{UNI}	ADD	5.5	0.95 (0.92-0.97)
	ABD	4.9	0.95 (0.91-0.97)

2.4. Statistical analysis

To identify significant differences in maximal ADD and ABD force, ADD:ABD, inter-limb asymmetries, and normalised muscle activity between assessment positions, multiple linear mixed models were performed. Levene's test of homogeneity were initially assessed, with the robust standard error correction applied if the assumption of homogeneity of variance was violated. For analysis, the assessment positions were set as the fixed effects, and the athletes were set as the random effect to account for the multiple measures recorded from the same athlete across positions. To determine whether there was a main effect of assessment position for all measures, Type III analyses of variance with the Satterthwaite approximation method for denominator degrees of freedom were used. Where a statistically significant main effect was identified, the estimated marginal means were calculated with *Bonferroni* post hoc analyses to determine the within assessment position

Table 2. Mean ± standard deviation of the hip adduction (ADD) and abduction (ABD) relative peak force, relative torque, ADD to ABD strength ratio (ADD:ABD), and the inter-limb asymmetry using the symmetry angle (SA) calculation, between the five assessment positions. *SE = seated; $SK_{0°}$ = supine at the knees with 0° hip and knee flexion; $SK_{45°}$ = supine at the knees with 45° hip flexion; SA_{BI} = supine at the ankles bilaterally; SA_{UNI} = supine at ankle unilaterally.*

	Relative Peak Force ($N{\cdot}BW^{-1}$)		Relative Torque ($Nm{\cdot}BW^{-1}$)			Asymmetry		
Assessment Position	ADD	ABD	ADD	ABD	ADD:ABD	ADD SA	ABD SA	ADD:ABD SA
SE	$0.51 \pm 0.09^{c,d}$	$0.50 \pm 0.08^{a,c,d}$	$0.30 \pm 0.12^{c,d}$	$0.29 \pm 0.09^{a,c,d}$	1.04 ± 0.21	1.2 ± 0.8^{d}	1.6 ± 1.1	2.2 ± 1.5^{d}
$SK_{0°}$	$0.47 \pm 0.13^{b,c,d}$	$0.39 \pm 0.10^{b,c,d}$	$0.28 \pm 0.13^{c,d}$	$0.23 \pm 0.09^{b,d}$	$1.22 \pm 0.31^{b,c}$	1.7 ± 1.5^{d}	1.7 ± 1.4	2.3 ± 1.9^{d}
$SK_{45°}$	$0.52 \pm 0.12^{c,d}$	$0.49 \pm 0.08^{c,d}$	$0.31 \pm 0.13^{c,d}$	$0.29 \pm 0.09^{c,d}$	1.06 ± 0.23	1.5 ± 1.4^{d}	1.8 ± 1.3	2.6 ± 2.2
SA_{BI}	0.23 ± 0.04	0.23 ± 0.03^{d}	0.21 ± 0.04	0.21 ± 0.03^{d}	1.01 ± 0.17^{d}	1.5 ± 1.7^{d}	1.2 ± 1.1	2.0 ± 2.3^{d}
SA_{UNI}	0.20 ± 0.04	0.18 ± 0.03	0.19 ± 0.04	0.16 ± 0.03	1.16 ± 0.14	3.4 ± 3.3	2.5 ± 2.0	4.7 ± 3.6

[a]significantly different to $SK_{0°}$; [b] significantly to $SK_{45°}$; [c] significantly different to SA_{BI}; [d] significantly different to SA_{UNI}.

differences. Additionally, Cohen's *d* effect sizes (ES) with 95% confidence intervals (95%CI) were calculated to infer the magnitude of within assessment position difference, with the following thresholds used; <0.20 (trivial), 0.20–0.60 (small), 0.61–1.20 (moderate), 1.21–2.00 (large) and >2.00 (very large) (Hopkins, 2002). All analyses were conducted in R Studio (v2021.09.1), with statistical significance set at $p \leq 0.05$.

3. Results

The mean and standard deviation (SD) for ADD and ABD rPF and rT, ADD:ABD, and the inter-limb asymmetries for each assessment position are outlined in Table 2. A significant main effect for assessment position was identified for ADD and ABD rPF ($F_{(4,76)} = 145.7$, $p < 0.01$; $F_{(4,76)} = 192.8$, $p < 0.01$, respectively), ADD and ABD rT ($F_{(4,76)} = 13.2$, $p < 0.01$; $F_{(4,76)} = 23.3$, $p < 0.01$, respectively), ADD:ABD ($F_{(4,76)} = 6.1$, $p < 0.01$), as well as ADD and ADD:ABD SA ($F_{(4,76)} = 4.6$, $p < 0.01$; $F_{(4,76)} = 4.5$, $p < 0.01$, respectively). No significant main effect for assessment position was determined for ABD SA ($F_{(4,95)} = 2.1$, $p = 0.09$). The ES comparisons between assessment positions for ADD and ABD rT, ADD:ABD, and ADD SA are presented in Figure 2.

For the RASUT, the mean (±SD) $\%EMG_{max}$ for each muscle was as follows: 100 ± 0% (REC_{ABD}), 36 ± 15% (ADD_{LONG}), 41 ± 18% (GRAC), 15 ± 9% (G_{MED}) and 16 ± 8% (SAR). Furthermore, the mean (±SD) $\%EMG_{max}$ for each muscle in the assessment positions are presented separately for ADD (Table 3) and ABD (Table 4). Significant main effects were identified between assessment positions during ADD for ADD_{LONG} ($F_{(5,95)} = 25.4$, $p < 0.01$), GRAC ($F_{(5,95)} = 65.7$, $p < 0.01$), SAR ($F_{(5,95)} = 15.8$, $p <$

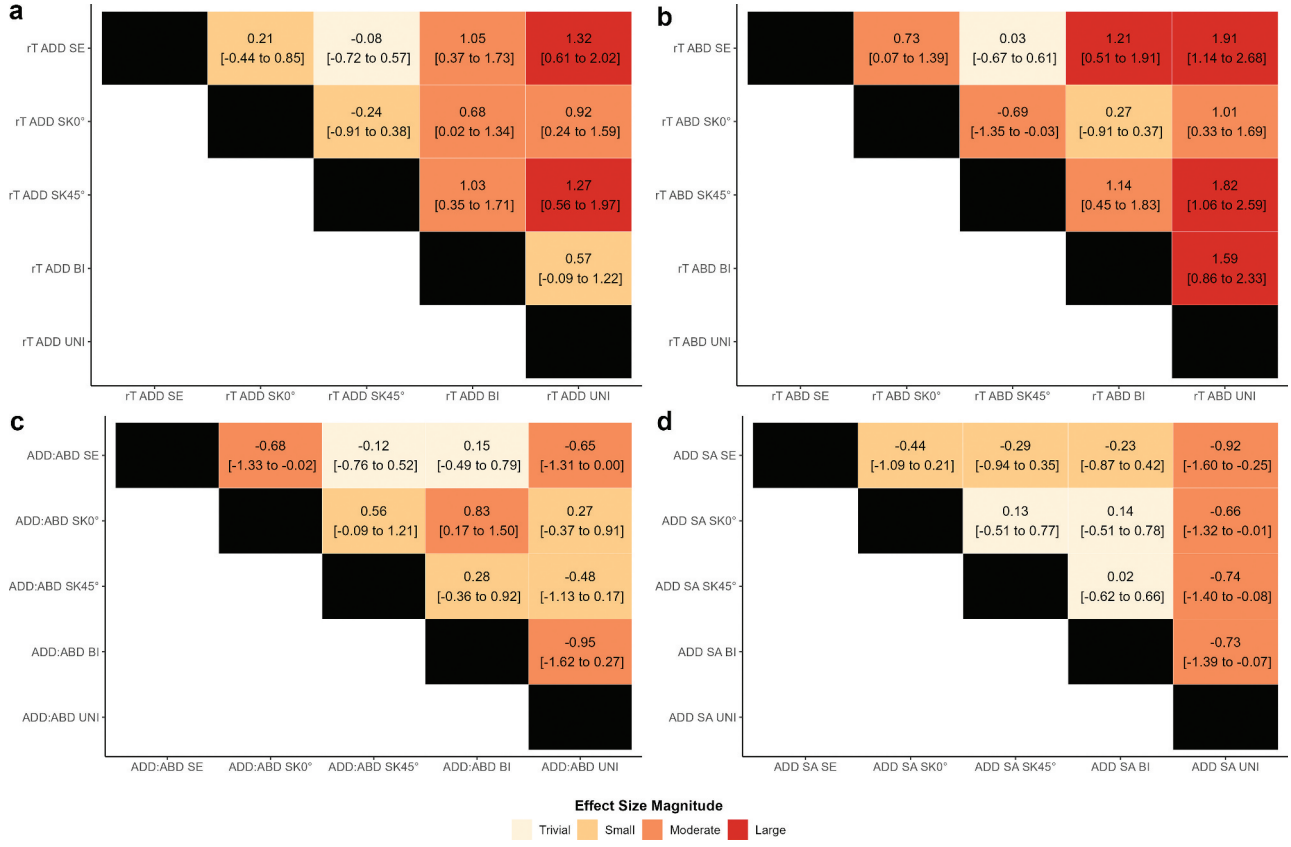

Figure 2. Cohen's *d* effect sizes (ES) [with 95% confidence intervals] for relative torque (rT) for; (a) adduction [ADD], (b) abduction [ABD], (c) the ADD to ABD strength ratio [ADD:ABD], and (d) the ADD symmetry angle [SA] between the five assessment positions. *SE = seated; SK0° = supine at the knees with 0° hip and knee flexion; SK45° = supine at the knees with 45° hip flexion; BI = supine at the ankles bilaterally; UNI = supine at ankle unilaterally. To interpret the figures read as the assessment position on the y-axis relative to the assessment position on the x-axis. For example, a negative ES means that the value for assessment position on the y-axis was lower than the value on the assessment position on the x-axis.*

Table 3. Comparison of the mean (±standard deviation) normalised maximal muscle activity (%EMG_{max}) achieved between the assessment positions during adduction (ADD). *SE = seated; $SK_{0°}$ = supine at the knees with 0° hip and knee flexion; $SK_{45°}$ = supine at the knees with 45° hip flexion; SA_{BI} = supine at the ankles bilaterally; SA_{UNI} = supine at ankle unilaterally.*

	Lower Rectus Abdominis (%EMG_{max})	Adductor Longus (%EMG_{max})	Gracilis (%EMG_{max})	Gluteus Medius (%EMG_{max})	Sartorius (%EMG_{max})
ADDUCTION ASSESSMENT POSITIONS					
SE ADD	16 ± 9	74 ± 22[a,e]	74 ± 14[a,b,d,e]	15 ± 11	19 ± 13[e]
$SK_{0°}$ ADD	16 ± 9	50 ± 18[b,c,e]	51 ± 19[b,c]	12 ± 8	10 ± 7[b,d,e]
$SK_{45°}$ ADD	14 ± 10[c]	86 ± 19[d,e]	91 ± 14[c,d,e]	13 ± 9	26 ± 13
SA_{BI} ADD	22 ± 15[d]	74 ± 20[e]	76 ± 15[d,e]	12 ± 7	17 ± 12[e]
SA_{UNI} (assessed leg)	13 ± 7	60 ± 16[e]	54 ± 16	12 ± 7	20 ± 13[e]
SA_{UNI} (support leg)	17 ± 11	32 ± 19	54 ± 21	13 ± 9	34 ± 17

[a]significantly different to $SK_{0°}$; [b] significantly to $SK_{45°}$; [c] significantly different to SA_{BI}; [d] significantly different to SA_{UNI} (assessed); [e] significantly different to SA_{UNI} (support).

Table 4. Comparison of the mean (±standard deviation) normalised maximal muscle activity (%EMG_{max}) achieved between the assessment positions during abduction (ABD). *SE = seated; $SK_{0°}$ = supine at the knees with 0° hip and knee flexion; $SK_{45°}$ = supine at the knees with 45° hip flexion; SA_{BI} = supine at the ankles bilaterally; SA_{UNI} = supine at ankle unilaterally.*

	Lower Rectus Abdominis (%EMG_{max})	Adductor Longus (%EMG_{max})	Gracilis (%EMG_{max})	Gluteus Medius (%EMG_{max})	Sartorius (%EMG_{max})
ABDUCTION ASSESSMENT POSITIONS					
SE ABD	18 ± 13	18 ± 9[e]	26 ± 13[e]	90 ± 11[a,c,d,e]	67 ± 16[b,e]
$SK_{0°}$ ABD	18 ± 16	17 ± 8[e]	23 ± 10[e]	55 ± 17[b,d,e]	73 ± 17[b,d,e]
$SK_{45°}$ ABD	17 ± 14	21 ± 10[e]	28 ± 12[e]	87 ± 10[c,d,e]	94 ± 9[c,d,e]
SA_{BI} ABD	22 ± 15[e]	17 ± 9[e]	22 ± 11[e]	62 ± 21[d,e]	63 ± 21[e]
SA_{UNI} (assessed leg)	18 ± 10	19 ± 7[e]	22 ± 9[e]	38 ± 20	58 ± 23[e]
SA_{UNI} (support leg)	14 ± 8	35 ± 14	44 ± 17	26 ± 18	21 ± 9

[a]significantly different to $SK_{0°}$; [b] significantly to $SK_{45°}$; [c] significantly different to SA_{BI}; [d] significantly different to SA_{UNI} (assessed leg); [e] significantly different to SA_{UNI} (support leg).

0.01), and REC_{ABD} ($F_{(5,95)} = 4.0$, $p < 0.01$), but not G_{MED} ($F_{(5,95)} = 1.8$, $p = 0.11$). Conversely, significant main effects for assessment position were observed in all muscles during ABD (G_{MED}: $F_{(5,95)} = 62.3$, $p < 0.01$; SAR: $F_{(5,95)} = 48.5$, $p < 0.01$; ADD_{LONG}: $F_{(5,95)} = 25.8$, $p < 0.01$; GRAC: $F_{(5,95)} = 21.2$, $p < 0.01$; and REC_{ABD}: $F_{(5,95)} = 2.3$, $p = 0.05$). Importantly, for all significant differences observed between assessment positions in the post hoc analyses for the ADD_{LONG} and GRAC in the ADD assessments the ES magnitudes were all deemed moderate to very large (ES = 1.02-2.86), and for the G_{MED} and SAR during the ABD assessments the differences were large to very large (ES = 1.54 to 3.98). Finally, during the ADD assessments small and moderate magnitude differences were observed for the REC_{ABD} between the SA_{BI} assessment position and the $SK_{45°}$ (ES = 0.59; 95%CI: 0.06 to 1.25) and SA_{UNI} assessed leg (ES = 0.76; 95%CI: 0.10 to 1.43) positions, respectively.

4. Discussion

The aim of this research was to compare the maximal hip ADD and ABD strength, ADD:ABD, inter-limb asymmetries, and normalised muscle activity in five commonly utilised isometric hip assessment positions. Significant main effects were identified for assessment position between all rPF, rT, ADD:ABD, and inter-limb asymmetry measures, except for ABD SA. The SE and $SK_{45°}$ positions elicited the greatest ADD and ABD rPF and rT, with moderate to large (ES = 1.03 to 1.91) significant differences compared to the SA_{BI} and SA_{UNI} positions. Additionally, rT for ABD, but not ADD, was significantly lower in the $SK_{0°}$ position compared to both the SE and $SK_{45°}$ positions, corresponding with a significantly higher ADD:ABD for $SK_{0°}$ compared to SE, $SK_{45°}$ and SA_{BI}. Moreover, the inter-limb asymmetries for ADD and ADD:ABD were significantly greater in the SA_{UNI} position compared to all others. Normalised maximal muscle activity differed significantly across assessment positions, with the $SK_{45°}$ position eliciting the highest %EMG_{max} for the ADD_{LONG} and GRAC during ADD assessments, as well as for the SAR during ABD assessments. Additionally, the G_{MED} exhibited the highest %EMG_{max} in the SE and $SK_{45°}$ positions for ABD, and the greatest REC_{ABD} %EMG_{max} was observed in the SA_{BI} position during ADD assessments.

The SE and $SK_{45°}$ positions exhibited significantly greater maximal relative isometric (rPF) and rotational (rT) ADD and ABD force compared to the other positions. Notably, during the ADD assessments, the $SK_{45°}$ position elicited the highest %EMG_{max} for the ADD_{LONG} and GRAC muscles (~86% and ~91%, respectively), followed by the SE (~74% and ~74%, respectively) and SA_{BI} (~74% and ~76%, respectively) positions. This data aligns with previous studies by Delahunt et al. (2017) and Lovell et al. (2012), which reported that ADD assessment positions in the mid-range (45°) of hip flexion produced significantly greater maximal ADD force and %EMG_{max} in the ADD_{LONG} and GRAC compared to positions with 0° and 90° of hip flexion. Interestingly, in this study the $SK_{45°}$ position also evoked the largest contribution from the SAR (~26%EMG_{max}) compared to the SE and SA_{BI} positions (~19% and 17%EMG_{max}, respectively). This data may explain the previous results of Light and Thorborg (2016), which reported that for bilateral short-lever ADD assessment positions in 45° of hip flexion, a position with the femurs placed in hip abduction produced significantly greater rT compared to that with a neutral femur position (0° of hip abduction). Therefore, it appears that the $SK_{45°}$ position on the rigid frame device may enable athletes to generate

maximal ADD force by allowing for the contribution of a medial hip rotation moment.

In the ABD assessments, significantly higher G_{MED} %EMG_{max} was observed in the $SK_{45°}$ (~87%) and SE (~90%) positions compared to the $SK_{0°}$, SA_{BI} and SA_{UNI} (~55%, ~62%, and ~38%, respectively) positions. Similarly, the SAR muscle demonstrated significantly greater %EMG_{max} in the $SK_{45°}$ position (~94%EMG_{max}) compared to all other positions. Interestingly, whilst the $SK_{0°}$ position demonstrated significantly lower ADD rPF (~10%) compared to the $SK_{45°}$ position, and significantly less ABD rPF (~22%) compared to both the SE and $SK_{45°}$ positions, these significant differences only remained for ABD and not ADD, when converted to maximal relative rotational force (rT). This aligns with the suggestion by McHugh et al. (2023) that practitioners should calculate and interpret rT for these assessments to adjust for lower-limb length differences, as the absence of these conversions may exaggerate hip ADD and ABD strength differences between athletes and the assessment positions. Furthermore, the significantly lower G_{MED} %EMG_{max} in the $SK_{0°}$ position compared to the SE and $SK_{45°}$ positions, suggests that the $SK_{0°}$ position places athletes at a mechanical disadvantage for producing maximal relative rotational force and eliciting maximal muscle activity. Importantly, previous research indicates that the G_{MED} experiences significantly greater excursion with hip flexion to 45° or 90°, compared to 0° (Jackson et al., 2023). This suggests that the $SK_{0°}$ position places the G_{MED} muscle fibres at a sub-optimal length to maximise force production (Ward et al., 2010). As such, this study highlights the suitability of the SE and $SK_{45°}$ assessment positions for practitioners seeking to assess maximal ADD and ABD strength, or monitor rehabilitation progress for injuries to the ADD_{LONG}, GRAC, SAR or G_{MED}.

Assessing isometric hip ADD and ABD strength in positions requiring force application at the knee (short-lever) is typically favoured in high-performance sport environments due to reduced athlete discomfort and a lower perceived risk of a muscular strain during testing. These positions also demonstrate high face validity to sports including mixed martial arts, wrestling and judo, where substantial ADD and ABD force production at the knees is crucial for manoeuvres such as takedowns, ground wrestling, and submission holds. Whilst the data on maximal force and muscle activity suggests that the SE and $SK_{45°}$ positions are the optimal short-lever assessments for screening and monitoring purposes, practitioners may still select the $SK_{0°}$ assessment position due to familiarity or the availability of population-specific reference values. As a result, it is noteworthy that the $SK_{0°}$ position elicited a significantly higher ADD:ABD (1.22 ± 0.31) compared to the SE and $SK_{45°}$ positions (1.04 ± 0.21 and 1.06 ± 0.23, respectively). Therefore, if utilising the $SK_{0°}$ assessment position, practitioners should consider applying a higher ADD:ABD threshold (<1.20) to identify potential "at-risk" athletes, as opposed to the previously reported threshold (<1.00) associated with groin injury risk (Tyler et al., 2001).

The importance of short-lever assessment positions for eliciting maximal ADD and ABD rotational force and %EMG_{max} in key muscles was highlighted in this study. However, long-lever assessment positions (SA_{BI} and SA_{UNI}) remain relevant in high-performance sport and rehabilitation settings due to their greater face validity and reported relationships with the mechanisms of non-contact lower-limb injuries and the performance of key technical activities, such as sprinting and change-of-direction in football codes and ice hockey (De Blaiser et al., 2021; Jones et al., 2021; Secomb et al., 2021). Specifically, significantly greater ADD SA and ADD:ABD SA were observed in the SA_{UNI} position compared to all other positions. This was consistent with previous research noting that unilateral strength assessments are the most sensitive for detecting inter-limb asymmetries (McHugh et al., 2023; Sarabon et al., 2020; Secomb et al., 2023). Given the previously identified relationship between greater ADD inter-limb asymmetry and an increased risk of future adductor strain injury, this assessment position will be important for practitioners in initial screening protocols (Markovic et al., 2020). Furthermore, this assessment position can also serve as a valuable tool for quantifying lower-body rehabilitation progress and informing targeted exercise prescription aimed at enhancing technical performance in sports requiring cyclical locomotion during sprinting and change-of-direction tasks, through reduced inter-limb asymmetries.

The majority of adductor injuries are attributed to the ADD_{LONG}, which along with the GRAC are reported to share common aponeuroses with the abdominal muscles, including the REC_{ABD} (Davis et al., 2012; Emblom et al., 2018; Schilders et al., 2017; Serner et al., 2015). Crucially, adductor strains with simultaneous abdominal-related injuries, and injuries involving disruption of the RA-adductor aponeurosis are associated with increased rehabilitation duration due to challenges in accurate diagnoses of the injury, and subsequent restoration of all relevant anatomical structures (Emblom et al., 2018; Hölmich et al., 2014; Thorborg et al., 2018). Therefore, assessments that require significant integration of adductor and abdominal muscle activity to produce maximal ADD force are essential for guiding rehabilitation from such injuries. Notably, during the ADD assessments, the SA_{BI} position elicited the largest %EMG_{max} in the REC_{ABD} (~22%) and was the second highest for ADD_{LONG} and GRAC %EMG_{max}. Similarly, Serner et al. (2014) compared various ADD exercises with yielding resistance, and the exercise performed in the same position as the SA_{BI} produced the greatest REC_{ABD} %EMG_{max} (~21%). Therefore, the SA_{BI} appears to be the most suitable assessment position for practitioners to monitor rehabilitation progress following an adductor strain or core muscle injury, as it elicited the greatest co-activation of the abdominal and adductor muscles.

A limitation of this study was that only a single athletic population was investigated. It is important to recognise that the physical demands of a sport and associated training practices may lead to specific adaptations in joint-angle specific strength and muscle activation patterns. This may result in varying differences observed between the assessment positions in other populations for the measures of force and muscle activity. Nevertheless, this study was the first to compare the maximal force output and normalised muscle activity in five commonly utilised hip ADD and ABD assessment positions, using a device consisting of a rigid frame fixated with individual load cells. This data offers valuable information to practitioners in selecting the most appropriate hip ADD and ABD assessment positions in high-performance

sport and sports medicine settings. Future research should aim to replicate this study with other populations including female athletes, and investigate whether low strength, greater asymmetries, or specific muscle activity patterns in the various assessment positions are associated with prospective lower-limb or abdominal muscle injury risk.

5. Conclusion

The results of this study highlight that significant differences existed between the hip ADD and ABD assessment positions for maximal force, ADD:ABD, inter-limb asymmetries and $\%EMG_{max}$ of all assessed muscles. This data has significant implications for practitioners selecting isometric hip ADD and ABD strength assessments for screening and monitoring purposes. It is anticipated that enhanced knowledge of the differences between these positions will allow practitioners to make better informed decisions when selecting the specific assessment position for their environment, or the individual athlete or client. Specifically, the assessment position selected should be the one which elicits an optimal response in the key measure of interest for the practitioner, whether that be determining maximal isometric or rotational ADD or ABD force, eliciting the greatest normalised activity capacity of specific muscles, evoking substantial co-activation of specific muscles, or identifying inter-limb asymmetries. This should ultimately enhance rehabilitation progress and the prescription of targeted exercise interventions aimed at reducing the opportunity for injury and improving technical performance, provided that the most appropriate test is selected, and the data is effectively interpreted.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The de-identified data associated with this study are available upon reasonable request.

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